

Counting extensions of `PerfectGroup(1920,1)` and `PerfectGroup(1920,2)` of order less than or equal to 122 880 elements

For either perfect group P , $N(n)$ is the number of $Z\varphi$ -classes, hence the number of isomorphism classes of extensions of P by solvable groups of order n . We write $C(n) = \sum_{1 \leq m \leq n} N(m)$.

1 ≤ n ≤ 32				33 ≤ n ≤ 64			
n	$ E = 1920n$	$N(n)$	$C(n)$	n	$ E = 1920n$	$N(n)$	$C(n)$
1	1 920	1	1	33	63 360	1	34 505
2	3 840	8	9	34	65 280	16	34 521
3	5 760	1	10	35	67 200	1	34 522
4	7 680	44	54	36	69 120	436	34 958
5	9 600	1	55	37	71 040	1	34 959
6	11 520	16	71	38	72 960	16	34 975
7	13 440	1	72	39	74 880	2	34 977
8	15 360	261	333	40	76 800	1 131	36 108
9	17 280	2	335	41	78 720	1	36 109
10	19 200	16	351	42	80 640	48	36 157
11	21 120	1	352	43	82 560	1	36 158
12	23 040	132	484	44	84 480	130	36 288
13	24 960	1	485	45	86 400	2	36 290
14	26 880	16	501	46	88 320	16	36 306
15	28 800	1	502	47	90 240	1	36 307
16	30 720	2 201	2 703	48	92 160	12 828	49 135
17	32 640	1	2 704	49	94 080	2	49 137
18	34 560	40	2 744	50	96 000	40	49 177
19	36 480	1	2 745	51	97 920	1	49 178
20	38 400	140	2 885	52	99 840	140	49 318
21	40 320	2	2 887	53	101 760	1	49 319
22	42 240	16	2 903	54	103 680	120	49 439
23	44 160	1	2 904	55	105 600	2	49 441
24	46 080	1 050	3 954	56	107 520	1 018	50 459
25	48 000	2	3 956	57	109 440	2	50 461
26	49 920	16	3 972	58	111 360	16	50 477
27	51 840	5	3 977	59	113 280	1	50 478
28	53 760	130	4 107	60	115 200	454	50 932
29	55 680	1	4 108	61	117 120	1	50 933
30	57 600	32	4 140	62	119 040	16	50 949
31	59 520	1	4 141	63	120 960	4	50 953
32	61 440	30 363	34 504	64	122 880	1 070 257	1 121 210

How to read the table. For $n = 48$, the table gives

$$N(48) = 12\,828, \quad |E| = 1920 \cdot 48 = 92\,160.$$

Therefore, for each of `PerfectGroup(1920,1)` and `PerfectGroup(1920,2)`, there are exactly 12 828 isomorphism classes of extensions by solvable groups of order 48; every resulting group has order 92 160. If the two distinct perfect residuals are counted together, this gives $2 \times 12\,828 = 25\,656$ groups of order 92 160.